

A Cooperative Game Theory Framework for Revenues Redistribution in Renewable Energy Communities

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Abstract

Renewable energy communities will be able to play a key role in the energy transition to climate neutrality. The development of energy communities requires an approach that combines technical, economic, and social expertise to address the many facets; in fact, energy communities aim to improve the use of locally generated renewable energy by promoting widespread community participation, thereby fostering environmental and social sustainability. Stakeholders include citizens, government agencies, and local businesses, facilitating the sharing of local energy production, stimulating the economy through local reinvestment of revenues, and reducing emissions by shortening the electricity supply chain. Many countries in the European Union, such as Italy, have identified a system of public incentives to attract all categories of stakeholders. At this time, the equitable distribution of economic benefits poses significant challenges, and cooperative game theory approaches can provide a technical solution that can improve the interest and number of participants involved. This paper examines the application of Shapley value, with an approach derived from cooperative game theory, for equitable benefit sharing among members of an energy community. A case study designed to fit the Italian scenario is presented to validate the applicability of the methodology. The framework evaluates each member's contribution proposing a scalable model for meritocratic revenue sharing that can improve the cooperation and efficiency of internal redistribution of energy community revenues.

1 Introduction

Renewable Energy Communities (RECs) have been introduced in the regulative framework of European Countries thanks to the Directive 2001/2018 (RED II) [1]. They represent one of the many faces of the energy transition toward the achievement of climate neutrality goals and require cross-cutting skills to deal with challenges that need to be addressed under different perspectives: technical, economic, and social. RECs aim to increase the percentage of renewable energy in the power grid by encouraging broad involvement of community members, thereby improving environmental and social sustainability. These partnerships can include different stakeholders, such as, for example, citizens, government agencies, local businesses, and industrial operators. RECs focus on enabling end users to share energy produced locally from renewable sources. By doing so they can enable the retention of value and resources within local areas. This can lead to a boost in the local economies, as revenues generated are captured and reinvested locally. Members of these initiatives are part of a unique cooperative system that is aligned with the principles of cooperative game theory, particularly with regard to the fair and equitable allocation of economic benefits. The application of game theory

principles can also be used to identify users' engagement policies.

Shapley value is used in cooperative game theory to determine the reward applicable to a player of a coalition, based on its marginal contribution. The same principle can be applied to identify benefit allocation policies (e.g., the more the user is valuable, the more benefits it will gain) as well as policies to actively engage users, thus generating, overall, a higher value for the whole coalition and ultimately result in a more effective and efficient utilization of energy.

In this scenario an energy community can be considered as a coalition in the sense of cooperative game theory. Scientific literature already advanced theoretical approaches that hypothesize the use of cooperative frameworks for optimizing operations of multi energy hubs (MEHs) [2]. Other researchers identified approaches that leverage sequential optimization [3] or mixed integer linear programming [4] adapted to a meritocratic redistribution framework. Similar approaches have been formulated and applied in configurations of projects realized in with co-investment strategies [5], [6].

The application of a cooperative game theory concepts to study the apportionment of incentives is not a nuance in the academic literature [7], [8], [9] yet the

methodology proposed by the authors is substantially different from already existing approaches as it aims at identifying the marginal contribution of each user under two different perspectives that pertain particularly (but not limited to) to the Italian context. These perspectives pertain to the ability of the user to contribute to the generation of incentive by injecting energy into the grid and the ability of the user to contribute to enhance the shared energy with its energy demand. This poses significant differences from existing work as it can provide essential information for both benefit sharing and the development of policies for the engagement and reward of users who change their habits to improve the overall performance of the configuration.

First, the theoretical framework is presented and adapted to fit Shapley's value to the unique dynamics of renewable energy communities. Then, a case study is proposed in which the developed framework is applied to evaluate the theorized approach. A comparative analysis is then conducted comparing a basic approach to revenue redistribution (proportional redistribution) with the one that exploits Shapley's value. The case study aims to demonstrate the benefits of using a cooperative game approach, instead of traditional ones, and that it can be applied in an industry relevant scenario to benefit multiple stages of the life cycle or a REC, from prototyping to management and active engagement.

2 Methodology

In cooperative game theory, the Shapley value stands as a fundamental concept for fairly distributing economic benefits among players involved in a coalition, providing a systematic approach to quantifying the marginal contribution of each player to the coalition's overall payoff.

Mathematically, the Shapley value for a player is calculated by considering all possible permutations of players joining the coalition and determining the marginal contribution of the player in each permutation. The average of these marginal contributions across all permutations yields the Shapley value for that player. The formula for calculating the Shapley value of a player i in a coalition N is expressed as:

$$\phi_i(N) = \frac{1}{|N|!} \sum_{S \subseteq N \setminus \{i\}} |S|! (|N| - |S| - 1)! (v(S \cup \{i\}) - v(S)) \quad (1)$$

Where $\phi_i(N)$ represents the Shapley value of player i in coalition N , $v(S)$ denotes the worth of coalition S , $|N|$ signifies the total number of players in coalition N and $|S|$ denotes the number of players in subset S .

In the devised framework, users have been evaluated under two perspectives: the first identifies their contribution in sharing energy by analyzing their injection into the grid, the second evaluates their ability to share energy by withdrawing it from the electrical grid. This has been applied specifically to the Italian scenario where an incentive tariff (TIP) is defined for the shared anergy. Such incentive depends on the market energy price, the power of the plant that injects energy into the grid and its hourly value can be calculated as follow:

$$TIP_h = \min[MaxValue; BaseValue + \max(0; 180 - Price_h)] \quad (2)$$

Specific details about the maximum value (MaxValue) and the base value (BaseValue) can be found in the operative rules published by the Italian authority [10].

1.1 Evaluation of contribution in generating incentive by injecting energy into the grid.

The methodology involves computing the marginal contribution to incentive generation for each participant and, subsequently, their Shapley value. This process entails the permutations of the participant in the coalitions to capture the unique contribution of each participant under various coalition scenarios. Then the marginal contribution is computed: for each permutation the shared energy and corresponding TIP has been incrementally calculated, thus assessing the contribution of each participant when added to the coalition. The marginal contribution is determined by the increase in total shared energy and TIPs due to a participant's inclusion. Finally, each participant's Shapley value is computed as the average of their marginal contributions across all permutations, providing a fair measure of their contribution to the REC's shared energy and economic benefits. Finally, the Shapley values that have been obtained are normalized to identify a "percentage of importance" for each user.

1.2 Evaluation of contribution in sharing energy by withdrawing

This second evaluation focuses on quantifying the contributions of each REC participant through its energy withdrawals.

First, the energy withdrawn which is valid for the shared energy is assessed: for each hour as the algorithm computes for the specific coalition of participants the total energy injected into the grid alongside the total energy withdrawn. Then it must be identified whether the shared energy is limited by the withdrawals or injections of energy.

Following this, the calculation focusses on the proportional energy withdrawal by each participant within the coalition. This aspect of the methodology involves a precise calculation where each participant's share of the total shared withdrawal (in kWh) is determined. The calculation considers their individual withdrawal requests in relation to the total withdrawal volume of the coalition or the total shared energy, depending on which is lesser. This nuanced approach ensures that the calculation accurately reflects each participant's capacity to withdraw against the backdrop of the total available shared energy, fostering a fair distribution mechanism.

Then the marginal contributions are computed evaluating the value of a coalition in terms of shared energy withdrawal both with and without the inclusion of a specific participant. Utilizing combinations and permutations of participants, the code iterates through every possible coalition, both including and excluding each participant, to assess their individual impact on the coalition's shared energy withdrawal capacity. Finally, the Shapley values are computed aggregating the marginal contributions of each participant across all possible coalitions they are part of. Ultimately, the obtained Shapley values are normalized for ease of comparison.

3 Case Study

To demonstrate the validity of the methodology a showcase of an energy community has been implemented. Fig. 1 shows the layout of the case study.

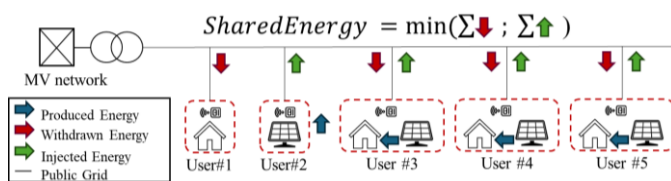


Fig. 1 Configuration of the case study.

The REC is composed of five members. The first one is a consumer, the second is a producer equipped with a 20 kWp photovoltaic plant, the other users are prosumers each with a 20 kWp PV plant.

A time horizon of 24 hours has been identified and used in the simulation. Fig. 2 shows the injection and withdrawal pattern for each user. The price of energy has been set constant to 100 €/MWh, even though the simulation framework allows to setup a different price for each hour.

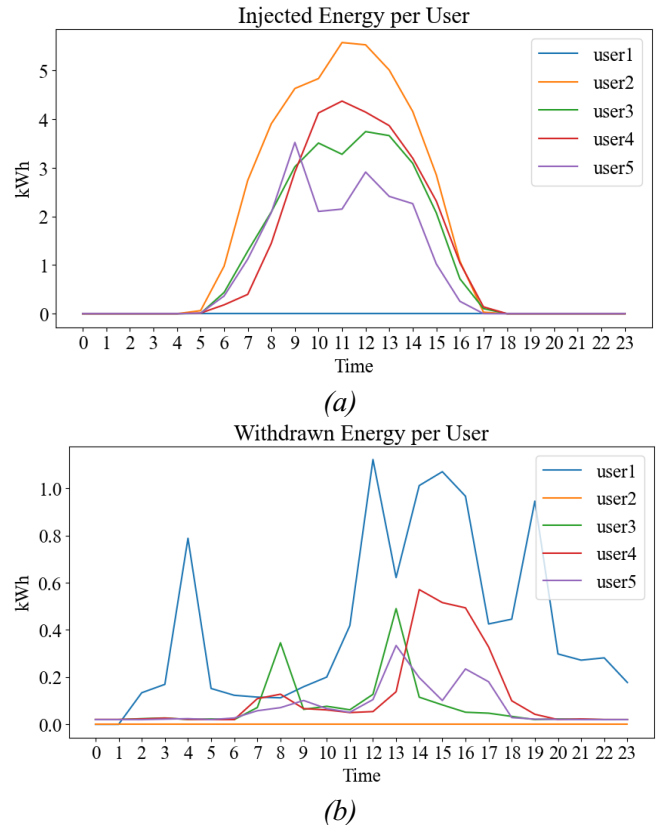


Fig. 2 (a) Energy injected during each period for each user, (b) Energy withdrawn during each period for each user.

The methodology previously described has been applied to the case study at first to identify the contribution of users to shared energy coming from the injection of electricity. Secondly, the contribution of each user to shared energy coming from withdrawal has been calculated. Shapley values results in table 1 have then been normalized to provide a more interpretable representation. It must be noted that for each configuration of the coalition the value of the shared energy changes accordingly.

Table 1 Shapley's values for the users

	User#1	User#2	User#3	User#4	User#5
Shapley injection	0.000	0.272	0.255	0.274	0.199
Shapley withdrawal	0.530	0.000	0.130	0.210	0.130

1.1. Discussion of results and comparison with other methodologies

Results show that User 1 does not contribute on generating incentive by injecting energy. The result is correct given that the user is a consumer. The same can be concluded for the contribution of User 2 by withdrawing energy given that the

user is a producer. In general, for energy injection User 2, User 3, and User 4 have similar Shapley values, with User 4 contributing slightly more than the others. This suggests that these users have similar levels of influence on the energy injected into the community's grid. For energy withdrawals, the Shapley's value of User 1 shows a significant impact on the REC: this is expected as it is the only consumer in the configuration which aids withdrawing energy at times when it was most valuable for balancing the grid's supply and demand. The contributions of User 3, User 4, and User 5 to withdrawal are notably less impactful compared to User 1 but suggest a beneficial consumption pattern that supports the REC's operational objectives. Table 2 shows the proportion of withdrawal and injection of the users in the configuration. This approach is an alternative more streamlined and direct to the marginal contribution approach.

Table 2 Proportional coefficients for the users

	User#1	User#2	User#3	User#4	User#5
Injection proportion	0.000	0.354	0.231	0.241	0.173
Withdrawal proportion	0.606	0.000	0.110	0.175	0.108

Comparing these results with those reported in Table 1 it is appreciable that, when it comes to benefit apportionment, Users 3, 4 and 5 generally are characterized by lower values; this would result in an unfair redistribution of benefits among users in favor of users 1 and 2 which, on the other hand, present significantly increased coefficients. This shows that, even a proportional approach is still a viable and valuable approach, it diminishes the actual contribution of some users to the configuration, in favor of other users thus not providing a righteous reflection of the real value of each user. Finally, it is worth noting that the software computes the Shapley value of all users for each hour of the day: with this information it is possible to give advices that aim at enhancing the overall performance of the community and increase users' contribution.

5 Conclusions

This paper delved into the equitable allocation of benefits in RECs by harnessing the Shapley value within a cooperative game theory framework. This work is motivated by the need for transparent and just reward mechanisms and to engage community members effectively. Methodologically, the paper introduced a dual-faceted evaluation approach, assessing both the energy injection and withdrawal patterns of users and determining their marginal contributions to shared energy. The case study applied this innovative framework, revealing that the Shapley value provides a more nuanced and equitable benefit distribution than a proportional methodology. Shapley value has the potential to benefit allocation in RECs, fostering a fairer, more cooperative community. Future developments are aimed towards the standardization of the

code, creating a less experimental version of it to facilitate its testing also in an operative environment.

5 Acknowledgements

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